Teaching Philosophy — Clear Fundamentals, Practical Wireless Engineering, and Inclusive Graduate Learning

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Teaching interests and experience

My teaching philosophy is grounded in the view that students learn engineering most effectively when they can explain underlying principles, implement a model, investigate real-world limitations, and communicate conclusions from evidence. In graduate teaching, this progression should be rigorous yet accessible: conceptual and mathematical foundations should lead naturally to simulation, critical evaluation, and informed engineering judgment.

I served as a Lecturer at the University of Lahore, teaching Wireless Communication, Communication Systems, and Computer Programming. I prepared lectures, assignments, examinations, and student-project guidance. At the University of Engineering and Technology, Taxila, I supported communications laboratories and experiments. At COMSATS University Islamabad, I advised student research activities involving simulation, experimental design, and technical documentation. These experiences demonstrated the importance of clear expectations, relevant examples, and constructive feedback when students bring different academic and professional backgrounds to the classroom.

My strongest teaching interest at Stevens is EE 583 Wireless Communications, which aligns directly with my academic teaching and doctoral research. I can contribute to topics such as wireless propagation and fading, link budgets, digital modulation, signal quality metrics, cellular architectures, and practical performance evaluation. I would also welcome teaching opportunities in communication systems, wireless networking, computational methods, and related ECE subjects as course needs arise. My Python/C/C++ and ML programming experience may inform discussion of CPE 810 GPU & Multicore Programming, but I would evaluate its detailed GPU programming requirements before accepting that specific course.

Instructional design and feedback

For EE 583, I would organize learning around a recurring engineering question: What does the model predict, how does the system behave when assumptions change, and how do we validate our answer? Students could compare analytical link-budget estimates with Python/MATLAB simulations, examine the effects of fading and interference, or evaluate simple modulation and channel-estimation methods under specified conditions. When appropriate, examples from LTE/5G network measurements would illustrate how measurements and implementation constraints differ from textbook idealizations.

Assessments would combine brief conceptual questions, quantitative problem sets, reproducible computational exercises, and one structured project. A sample project could ask teams to design and evaluate a wireless link across different environments, justify assumptions, and report both performance and limitations. Rubrics would explain criteria for mathematical reasoning, implementation, verification, interpretation, and communication. I would provide feedback early enough for students to revise mistakes, and use short formative checks to identify gaps in signal-processing prerequisites.

For graduate learners, I also value research literacy. Students should learn to compare methods against meaningful baselines, understand data provenance, and distinguish results that generalize from results that depend on particular simulation settings. My research in domain adaptation, synthetic-data quality, and operational reliability provides a useful context for these discussions without requiring every student to specialize in AI.

Inclusion, advising, and graduate mentoring

An inclusive graduate classroom recognizes that students enter with diverse preparation in programming, communications theory, and applied mathematics. I would make prerequisites explicit, offer short optional refreshers, and provide multiple routes into core ideas through equations, diagrams, code, and interpretation of measured data. Structured project milestones help students manage challenging assignments while protecting academic standards.

My classroom expectations emphasize mutual respect, accessible course materials, transparent grading, and opportunities to ask questions without embarrassment. I would offer predictable office hours, respond to students with timely, specific guidance, and make reasonable adjustments through established university accessibility procedures. Collaborative assignments would include clear team roles and individual accountability so that participation and assessment remain fair.

In graduate student advising and research mentoring, I use a staged process: first reproduce a baseline, then test a controlled variation, and finally formulate an independent question. This helps students develop confidence and technical judgment without compromising rigor. Academic integrity, responsible use of AI-assisted software, honest reporting of negative results, and respect for collaborators are central to the process.

Continuous improvement and departmental coordination

I would work with the Stevens ECE department to align the course with approved outcomes, prerequisites, and assessments, and report student progress or emerging challenges promptly. I would use assessment results and course feedback to refine examples, pacing, and assignments each term. My objective is to help Stevens students connect the principles of wireless communications with modern engineering systems, disciplined experimentation, and responsible professional practice.